<<S.M.H.Mousavi_Patent_Description>>

Patent Title: **Portable system for monitoring emotional state and concentration in interviews based on multi-modal fusion of visual and audio data on a single-board computer platform**.

- **Description of the Invention**

Title of the Invention

Portable System for Monitoring Emotional State and Focus During Interviews Based on Multi-Channel Audio-Visual Data Fusion on a Single-Board Computer Platform

- **Technical Field of the Invention**

This invention falls within the field of intelligent systems for non-contact monitoring of individuals' emotional and cognitive states. Leveraging image processing, audio processing, and multi-channel data fusion deployed on a single-board microcomputer hardware platform based on Raspberry Pi, the system analyzes human behavior during interview processes.

- **Technical Problem and Statement of Invention Objectives**

In interview processes, evaluating an individual's emotional, cognitive, and behavioral state typically relies on human judgment, direct observation, or subjective questionnaires. These conventional methods suffer from notable limitations, including bias from the evaluator's subjective viewpoint, human error, lack of temporal continuity in data collection, and an inability to simultaneously capture an individual's multi-channel physiological and behavioral reactions. Furthermore, many existing systems either depend on expensive, complex equipment or analyze only a single isolated data channel (such as video or audio independently) without offering structured fusion across multi-source data streams. Additionally, the lack of an integrated, non-contact system implementable on lightweight hardware has severely hindered practical deployment in real-world interview environments.

The present invention is introduced to overcome these limitations. It presents an intelligent non-contact monitoring system that simultaneously employs image processing, audio processing, and multi-channel data fusion to provide continuous, quantitative behavioral analysis during interviews. The primary objective is to extract and fuse multimodal behavioral indicators—including facial, ocular, oral, and bodily movements, acoustic speech patterns, and chest respiratory signals—transforming them into high-level emotional and cognitive indices.

Implemented on a Raspberry Pi single-board computer, the system reduces hardware cost and complexity, achieves real-time execution in real-world environments, and delivers an objective, accurate, and repeatable behavioral assessment throughout an interview.

- **Description of Prior Art and Related Technological Developments**

In the domain of human behavioral and emotion analysis, several patents and technical solutions exist that attempt to detect emotional states from visual and audio data; however, each presents distinct limitations compared to the solution proposed in this invention.

One related prior art reference is patent US20210390288A1, which discloses a method and system for detecting human emotion in images or video through image feature extraction for emotion classification. While this system detects emotional states via video-based facial expression analysis, it operates independently of audio analysis or multi-channel data fusion, focusing exclusively on visual inputs.

(Google Patents link: https://patents.google.com/patent/US20210390288A1)

Another relevant patent, US11830291B2, relates to a system and method for multimodal emotion recognition that attempts to combine audio, visual, and textual features within a machine learning model. This invention employs multi-channel fusion techniques to enhance detection accuracy; however, it remains confined to theoretical neural network training frameworks and fails to address practical real-world interview constraints or real-time execution on lightweight embedded hardware.

(Google Patents link: https://patents.google.com/patent/US11830291B2)

Additionally, Chinese patent CN117708752A addresses an emotion detection system based on video and audio data fusion, presenting feature extraction and integration techniques for emotion detection. Nevertheless, it lacks support for real-time edge processing and fails to provide high-level behavioral assessments customized for practical application scenarios.

(Google Patents link: https://patents.google.com/patent/CN117708752A)

Another patent describing an audio-visual fusion framework is CN114399818A, which discloses a multimodal facial emotion recognition method and apparatus. This patent concentrates primarily on algorithmic precision and the application of heavy deep learning models for face and voice analysis, offering minimal focus on operational behavioral management in real-life interviews.

(Google Patents link: https://patents.google.com/patent/CN114399818A)

In summary, although the aforementioned references pertain to the broad domain of emotion recognition from audio-visual data, none offer simultaneous multi-channel data analysis tailored for practical interview scenarios, real-time edge execution on lightweight single-board computers, or structured fusion targeting psychological and cognitive evaluation within a self-contained, integrated apparatus.

In contrast, the present invention delivers an intelligent, non-contact monitoring system for emotional and cognitive state assessment during interviews via concurrent audio-visual feature extraction and multi-channel fusion, embodied in an integrated hardware-software architecture deployable on lightweight platforms.

- **Detailed, Sufficient, and Integrated Description of the Technical Solution**

The present invention provides an intelligent, non-contact monitoring system for assessing emotional and cognitive states during interviews. In an integrated, real-time manner, it captures, processes, and analyzes multi-channel visual and auditory data to derive behavioral, emotional, and cognitive metrics. The system is engineered to operate without requiring physical contact, wearable sensors, or active subject cooperation, enabling seamless behavioral profiling under real-world interview settings.

As illustrated in the technical drawings, the apparatus comprises a central processing unit based on a Raspberry Pi board (Figure 4, component 3), interfaced with:

A color capture camera module (12) operating at 640×480 resolution and approximately 15 frames per second;

An acoustic capture microphone (11);

A 7-inch touchscreen display (2);

A mainboard power supply (4);

An available power level indicator (20);

A wireless mini keyboard with integrated touchpad (13).

Notably, the camera module is mounted with multi-directional pan-tilt articulation. To optimize throughput and bypass USB bus latency, a Camera Serial Interface (CSI) camera module is utilized, connecting directly to the Raspberry Pi mainboard ribbon interface (9). All computation is conducted strictly locally and edge-based, completely independent of cloud infrastructure. The processing board utilizes a 4th-generation Raspberry Pi (19). For convenient touchscreen interaction, a dedicated stylus pen (1) is provided. The Raspberry Pi board incorporates two USB 3.0 ports (5) and two USB 2.0 ports (6), utilized for interfacing the

microphone, wireless keyboard RF receiver, and a 5V LED illumination module (10). Power is supplied through a USB-C port (7), while display output is routed via an HDMI cable to the Mini HDMI port (8). Active thermal management for CPU/GPU cooling is driven via GPIO pins 4 and 6 (18).

Due to the absence of suitable commercial off-the-shelf enclosures, the comprehensive device chassis (17) was custom-fabricated via 72 hours of continuous 3D printing. An ergonomic carrying handle (14) is seamlessly molded into the main enclosure, and an elevated support stand (15) is integrated into the base assembly to guarantee an optimal line of sight and operational angle for the interviewer.

- **Visual and Auditory Processing Pipeline:**

Facial Detection and Localization: The live video stream captured by the camera is continuously converted to grayscale. Utilizing a Haar Cascade classifier, the primary facial bounding box is localized. When multiple subjects appear, the algorithm isolates the largest bounding box as the active target, establishing the baseline region of interest (ROI) for all downstream visual analyses.

Ocular Feature Extraction: The upper half of the detected facial ROI is defined as the ocular region. A secondary cascade classifier isolates the eye coordinates. The illumination intensity across each eye ROI is computed from mean pixel values and normalized to a [0, 1] range. The mean brightness across both eyes serves as the ocular intensity index, while its inverse reflects ocular activity level (where rapid drops in brightness correspond to blinking or eye closure events). Furthermore, spatial intensity distributions across horizontal and vertical axes determine the relative gaze coordinates, classifying gaze direction into distinct qualitative categories: Center, Left, Right, Up, or Down.

Oral Movement and Speech Dynamic Analysis: A relative bounding box spanning the lower third of the facial ROI is extracted to monitor mouth dynamics. Following Gaussian smoothing and adaptive thresholding, mouth aperture is quantified via vertical pixel variance. Inter-frame fluctuations in this value define the raw mouth activity metric. This signal is stabilized using an Exponential Moving Average (EMA) filter to dampen transient sensor noise and isolate authentic speech articulation. Furthermore, mouth activity is adaptively cross-referenced with acoustic energy to eliminate non-verbal oral motions (e.g., yawning or chewing).

Respiration and Body Motion Extraction: A relative sub-region spanning the lower jaw down to the upper torso is tracked across consecutive frames using Optical Flow algorithms. The average optical flow vector magnitude across this region serves as the raw chest motion metric.

Normalized to a [0, 1] scale, this metric reflects physical movement intensity and respiratory rhythm. All computer vision routines are implemented natively using OpenCV.

Acoustic Signal Processing: In parallel, ambient acoustic signals are digitized via the microphone at a 16 kHz sampling rate. Over short temporal analysis windows, acoustic features are extracted: Root-Mean-Square (RMS) Energy, Peak Amplitude, Zero-Crossing Rate (ZCR), and Fundamental Speech Frequency (Pitch) via Autocorrelation Analysis. Normalized acoustic energy represents conversational vocal intensity. To accommodate the edge computational budget of the Raspberry Pi and ensure deterministic real-time execution across numerous data channels, heavy deep learning models are avoided in favor of lightweight, mathematically rigorous feature extraction algorithms.

- **Multi-Channel Fusion and High-Level Behavioral Indices:**

In the multi-channel fusion layer, normalized features derived from oral dynamics, ocular tracking, gaze vector, chest displacement, and acoustic signals are combined using an adaptive weighting matrix:

Stress Index: Computed as a weighted composite of mouth activity, respiratory/chest displacement, acoustic energy, ocular blink rate, and gaze deviation.

Focus Index: Derived from gaze vector stability, reduced blink/ocular restlessness, and minimized erratic torso movement.

Confidence Index: Calculated inversely relative to instantaneous stress and directly proportional to sustained focus and stable vocal pitch.

(Channel specifications and weighting parameters are detailed in Table 1).

Based on these fused vectors, the subject's instantaneous emotional state is mapped into discrete, interpretable states: Anxious, High Stress, Focused, Confident, Engaged, Distracted, or Neutral. This rule-based classification ensures transparent interpretability without relying on opaque, high-latency neural network inference.

Beyond instantaneous emotional states, the system aggregates time-series metrics into six high-level professional assessment indicators displayed in real-time:

1. Trustworthiness
2. Employability
3. Mental Resilience
4. Social Dominance / Presence
5. Psychological Fatigue

6. Behavioral Adaptability

All temporal data, raw extracted features, dimensional reductions, and final evaluation summaries are logged automatically into structured text and tabular formats according to Table 2, providing verifiable audit trails. The system operates continuously throughout the interview until gracefully terminated by user input (keyboard key “q”). The overall architectural workflow is illustrated in the flowchart in Figure 12.

Consequently, the invention provides a low-cost, non-contact, portable edge apparatus engineered specifically for practical interview evaluation, resolving prior art shortcomings by executing localized, real-time multi-channel behavioral analysis.

- **Description of Drawings and Figures**

Figure 1: Front perspective view of the portable system, illustrating input sensor arrangements and status/illumination output LEDs.

Figure 2: Lateral profile view of the apparatus, detailing the relative mechanical mounting of the display, core chassis, and user input hardware.

Figure 3: Opposing lateral view of the system, illustrating touchscreen mounting, stylus cradle, and external peripheral interfaces.

Figure 4: Detailed hardware view of the 4th-generation Raspberry Pi single-board computer, indicating CPU/GPU, memory modules, I/O headers, USB-C power interface, GPIO pins, USB ports, micro-HDMI ports, and the dedicated CSI camera ribbon connector.

Figure 5: Color camera module assembly with flexible CSI flat cable, utilized for real-time facial video capture.

Figure 6: Acoustic transducer (microphone) assembly, employed for recording ambient and speech acoustic signals during interview workflows.

Figure 7: System power management module with integrated charge level indicator, providing uninterrupted regulated electrical power.

Figure 8: 7-inch integrated capacitive touchscreen display, displaying real-time analytical metrics, raw telemetry, and operational states.

Figure 9: Stylus accessory engineered for precision touchscreen navigation.

Figure 10: Ultra-compact wireless keyboard and capacitive touchpad controller, facilitating operator input and diagnostic overrides.

Figure 11: 5V auxiliary LED illuminator module, ensuring calibrated, shadow-free facial illumination in substandard ambient lighting environments.

- **Distinct Technical Advantages Over Prior Art**

Completely Non-Contact Architecture: Eliminates wearables, skin-contact electrodes, or intrusive physiological probes, avoiding Hawthorne effects and preserving natural subject behavior under real-world interview stress.

Synchronized Multimodal Fusion: Rather than relying on isolated voice or facial models, it establishes a mathematically synchronized multi-channel pipeline incorporating facial expressions, micro-blinking, gaze vectors, gross body motion, respiratory chest displacement, and acoustic speech mechanics.

Application-Specific Design for Human Interviews: Unlike generic emotion classifiers, the metric weights are calibrated to quantify professional dynamics: stress tolerance, conversational engagement, cognitive fatigue, and communicative confidence.

Cloud-Independent Real-Time Edge Processing: Operates self-contained on a low-power single-board computer without network dependencies, guaranteeing data privacy, zero network latency, and field deployment in secure facilities.

Algorithmic Efficiency and Deterministic Execution: Replaces computationally heavy deep networks with lightweight, optimized classical feature extractors, maintaining high frame rates (11–15 FPS) on edge silicon.

Longitudinal Trend Logging: Provides automated structured exports of frame-by-frame and aggregated indices for subsequent review, comparative candidate ranking, and forensic analysis.

- **Detailed Operating Procedures and Use Cases**

1. Primary Operating Procedure (Interview Scenario)

Consent and Protocol Briefing: The subject is informed of the privacy-preserving data policy: raw biometric video and voice recordings are not stored; only anonymized, numerical mathematical features are logged into tabular formats strictly for evaluation purposes. The subject retains the right to withdraw at any time.

System Initialization: Power is toggled via the lateral power bank switch, initiating the Raspberry Pi OS. The operator opens the Thonny IDE environment and launches the behavioral analysis application via the Run command (green arrow).

Environmental Setup: If ambient lighting is deficient, the 5V LED illuminator is switched on. The interviewee is seated 1.0 to 3.0 meters directly facing the CSI camera lens.

Autonomous Tracking: The system autonomously localizes the face, binds the target tracking ROI, and begins real-time processing without operator intervention.

Synchronized Extraction & Telemetry: Ocular blinks, gaze vectors, oral articulation, optical-flow chest breathing dynamics, and speech prosody are continuously sampled. The multi-channel fusion layer calculates real-time stress, focus, and confidence curves, alongside the six cumulative behavioral indices on the 7-inch display.

Graceful Shutdown: The session terminates upon pressing the “q” key. Data files are finalized and stored to non-volatile storage. The operator initiates software shutdown (Shut Down), followed by a double-click on the power module switch to disconnect power rails completely.

2. Educational Classroom Monitoring Use Case

The device is positioned facing student desks. Following identical privacy disclosures and initialization, the unit tracks learner focus, cognitive engagement, and restlessness over academic lectures, logging aggregate behavioral engagement without recording student likenesses.

3. Athletic and Sports Performance Use Case

Positioned during training drills, the apparatus monitors an athlete’s physical breathing frequency, gaze fixation under exertion, and psychological fatigue, displaying fatigue trends on-screen to prevent overtraining and optimize cognitive resilience.

- **Explicit Industrial and Commercial Applications**

Human Resources and Talent Acquisition: Objective, bias-free candidate screening during structured corporate interviews.

Educational Institutions and Exam Proctoring: Real-time engagement quantification and attentiveness auditing during instructional sessions.

Athletic and Sports Science Centers: Non-invasive monitoring of mental stamina, exertion-related respiratory patterns, and focus under physical strain.

Psychological and Clinical Behavioral Research: Quantitative, privacy-compliant tracking of emotional states, anxiety indicators, and conversational dynamics.

Remote Learning and Video Conferencing Systems: Embedded telemetric feedback systems evaluating participant engagement during virtual training modules.